ML 1.2

Saturday, February 17, 2024 9:06 PM

Intro to ML and NN

1.0

↳ Linear Regression ←

Logistic Regression ←

{ Evaluate

Train Validation Test

- overfit
- underfit

→ Regularization

L_1 L_2

Linear Regression

{ RSS, R^2, MSE }

Logistic Regression

↓ classification

- Accuracy

Imbalanced

Precision FP ↓

Recall FN ↓

F1 score

Threshold ≠ ROC AUC IPR / FPR

Imbalanced

↓

P R (Precision, Recall) } curve

Imbalance

under over SMOTE

Confusion - { 0 , 1

{ PCA } 7 ~~→~~ 2 Visual

* yes - Logistic but its is of

No use

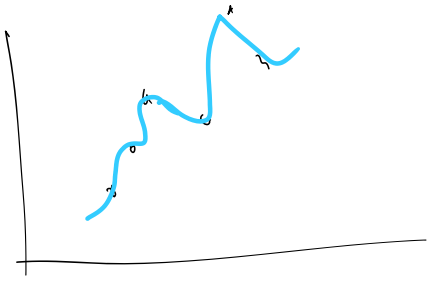
↓

Log loss = { $w^T x + w_0$

Linearly separeal

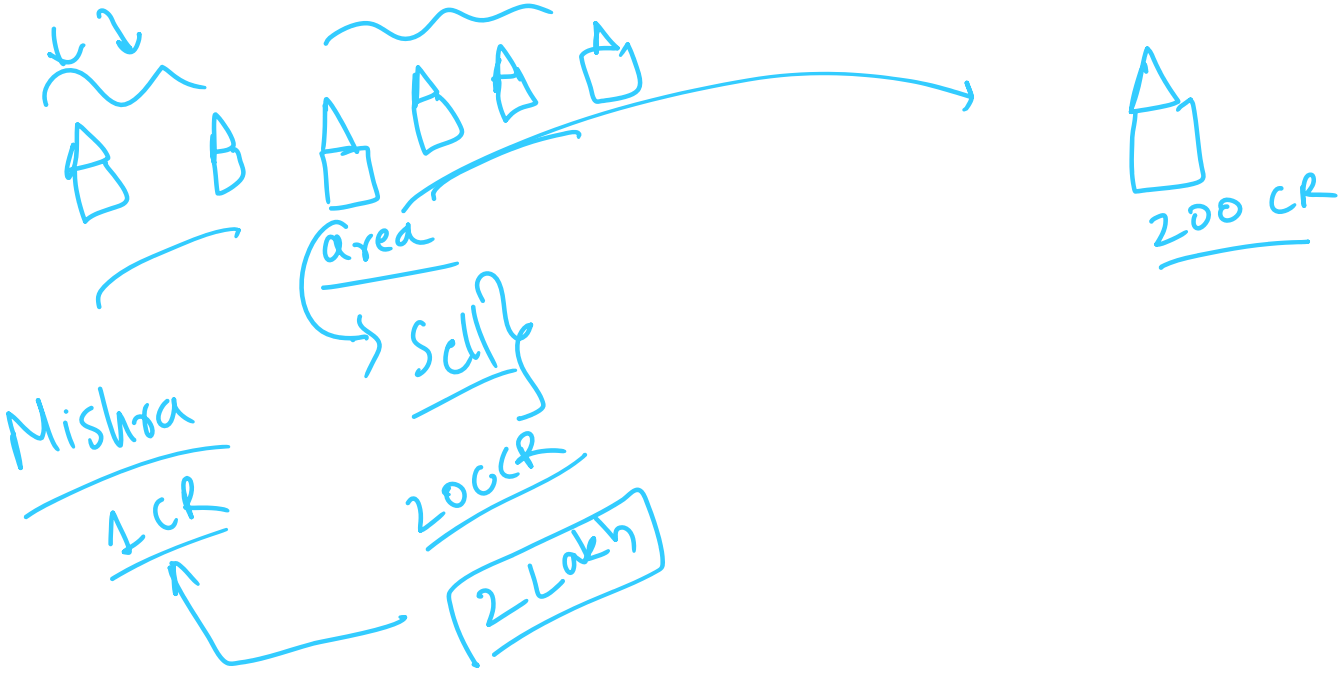
Linearly separable

- Polynomial → overfit
- Multiclass → OVR

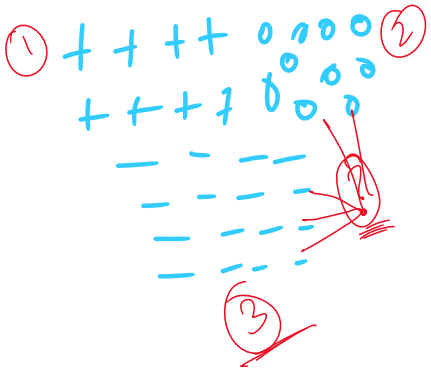


overfit

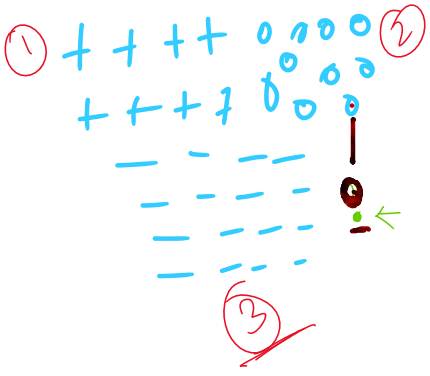
Linear



K Nearest Neighbor

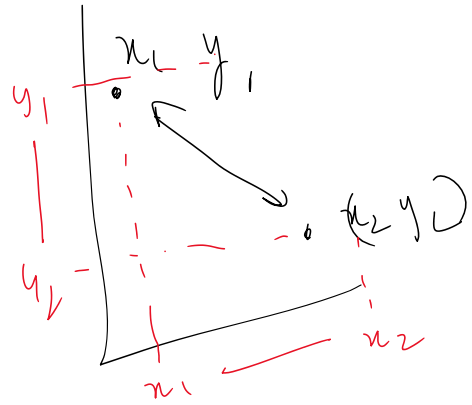


	f_1 Marks Eng	f_2 Math	f_3 Bio	Class
	92	93	94	Topper
	91	94.5	96	
	60	30	40	Class 2



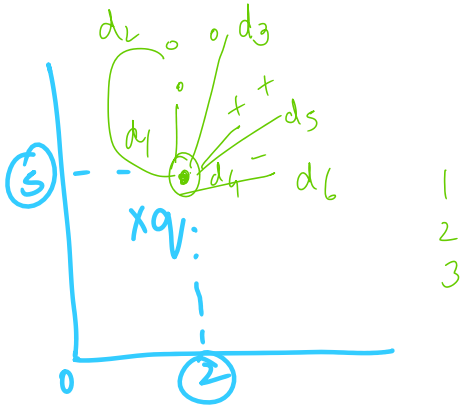


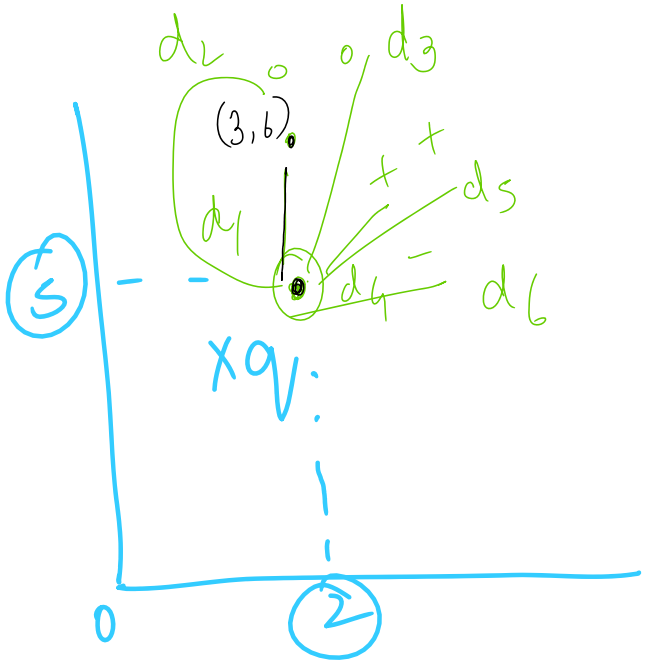
distance



$$\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$





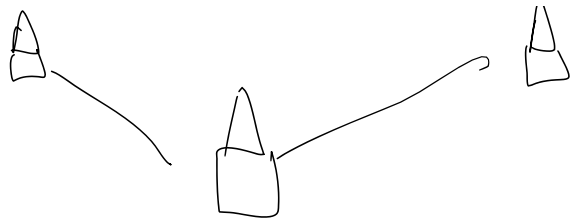


$$\sqrt{(2-3)^2 + (5-6)^2}$$

$$\sqrt{1^2 + 1^2}$$

$$\sqrt{2}$$

$$1.41$$



Nearest — Shortest distance

{ 3, 5, ≠, 9 } Hyperparameter

K

K — N Nearest N Neighbor

→ top 3
→ top 5

Logistic

Weights

$\{w_1 \quad w_2 \quad w_0\}$

Learn

Parametric

KNN

No weights

d_1

Take the shortest

Parametric

the Shortest

Non Parametric

d_1 — 1

s1- find majority vote
s2- perform euclidean distance
s3- sort and select k datapoints
s4- give class to x_q datapoint

From <https://www.scaler.com/instructor/meetings/i/ml-k-nearest-neighbors-1-9/>

S_2 — dist

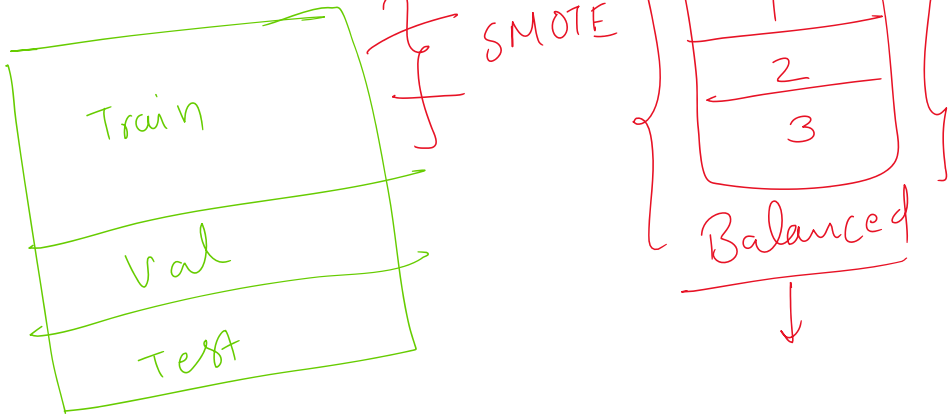
S_3 — 3 Top

S_1 — vote

S_4 →

Algo — understand
Math
code

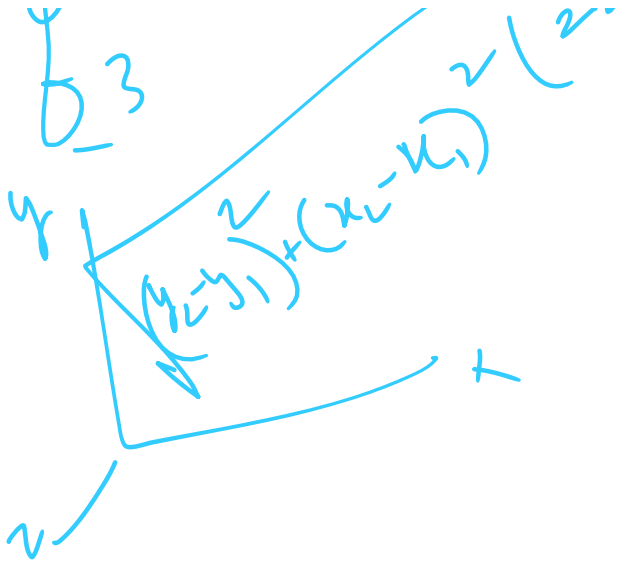
} Limitation
Assumption



f_1 f_2 f_3

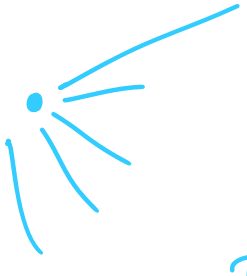
$2^{2^{2^{2^k}}}$

y_1 y_2 y_3



{ Test — 110 Datapoint

Train + val
300 Points }



300 * 110

Distance →

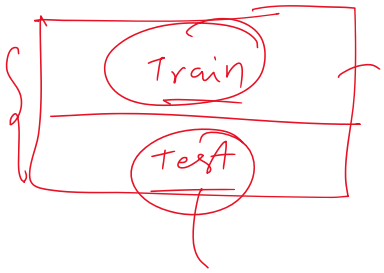
NN — RNN → Transformers → LLM

{ weighted KNN }

2
7
3
10
11
12

$\frac{x - x_{min}}{x_{max} - x_{min}}$

Max
min



fit transform

Unseen

